

Heston Model Simulation and Theoretical RND

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July 27, 2024

The Heston model is a popular stochastic volatility model used in financial mathematics for option pricing. It was introduced by Steven Heston in 1993 Heston (1993). The model is designed to capture the empirical observations of financial markets, such as the volatility smile, by allowing the volatility of an asset to be stochastic (i.e., changing over time).

Heston Model Overview

The Heston model describes the dynamics of the asset price S_t and its variance v_t under the risk-neutral measure $\mathbb{Q}$ using the following system of stochastic differential equations (SDEs):

$$\begin{aligned}dS_t &= \mu S_t dt + \sqrt{v_t} S_t dW_t^S, \\dv_t &= \kappa(\theta - v_t) dt + \sigma \sqrt{v_t} dW_t^v,\end{aligned}$$

where:

- S_t is the price of the underlying asset at time t .
- v_t is the instantaneous variance of the asset price.
- μ is the drift rate of the asset price under the risk-neutral measure.
- κ is the rate of mean reversion of the variance.
- θ is the long-term mean variance.
- σ is the volatility of variance, often referred to as the “vol of vol”.
- W_t^S and W_t^v are two Brownian motions with correlation ρ .

Risk-Neutral Density

The risk-neutral density of the underlying asset price at a future time T is crucial for pricing derivative securities. To derive the risk-neutral density $p(S_T|S_0)$ from the Heston model, we typically follow these steps Heston (1993):

Characteristic Function

The first step involves deriving the characteristic function of the log-price $\ln(S_T)$. The characteristic function $\phi(u)$ is defined as:

$$\phi(u; S_0, v_0, t, T) = \mathbb{E}^{\mathbb{Q}} \left[e^{iu \ln(S_T)} \mid S_0, v_0 \right],$$

where i is the imaginary unit.

The characteristic function in the Heston model can be expressed in a semi-closed form. The key is to solve the Riccati differential equations that arise from the model's SDEs. The characteristic function $\phi(u)$ of $\ln(S_T)$ is given by:

$$\phi(u) = \exp \left(C(T - t, u) + D(T - t, u)v_0 + iu \ln(S_0) \right),$$

where $C(\tau, u)$ and $D(\tau, u)$ solve the following Riccati equations:

$$\begin{aligned} \frac{\partial D(\tau, u)}{\partial \tau} &= \kappa \rho \sigma i u D(\tau, u) + \frac{1}{2} \sigma^2 D(\tau, u)^2 + i u, \\ \frac{\partial C(\tau, u)}{\partial \tau} &= \kappa \theta D(\tau, u). \end{aligned}$$

with terminal conditions $D(0, u) = 0$ and $C(0, u) = 0$.

Fourier Transform

The risk-neutral density $p(S_T|S_0)$ can be obtained by taking the inverse Fourier transform of the characteristic function. This step leverages the fact that the density function and the characteristic function are Fourier transform pairs.

To recover the risk-neutral density $p(S_T \mid S_0)$, we use the inverse Fourier transform of the

characteristic function. The density function is given by:

$$p(S_T | S_0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iu \ln(S_T)} \phi(u) du. \quad (1)$$

In practice, this integral is often computed numerically. Additionally, since the Heston model deals with log-prices, we often express the density in terms of $x = \ln(S_T)$:

$$p(x | \ln(S_0)) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iux} \phi(u) du. \quad (2)$$

Practical Implementation

To compute the theoretical risk-neutral density using the Heston model, we typically follow these steps:

1. Calculate the Characteristic Function: Solve the Riccati equations to obtain the characteristic function $\phi(u)$.
2. Numerical Inversion: Use numerical methods such as the Fast Fourier Transform (FFT) to invert the characteristic function and obtain the risk-neutral density.

Summary

The Heston model is a sophisticated tool for capturing the dynamics of asset prices with stochastic volatility. To compute the theoretical risk-neutral density of the underlying asset price, we derive the characteristic function of the log-price and use the inverse Fourier transform. This allows us to obtain the Risk-Neutral Density RND, which is essential for option pricing and other financial calculations.

Experiments

First, we simulate N options using the Heston model, where N is chosen to be a large number. Subsequently, we compute the theoretical risk-neutral density (RND) p_H based on the analytical expression in Equation 2.

Random Sampling k options

We sample k options from the N simulated options with replacement, where $k \ll N$. Using methods such as CGP and others, we then estimate the risk-neutral densities p_{CGP} , p_{GP} , and so forth. To compare the theoretical RND p_H with the estimated RNDs, we employ various comparison methods. These methods include Euclidean Distance, Kullback-Leibler (KL) Divergence Kullback and Leibler (1951), Jensen-Shannon (JS) Divergence Tishby et al. (2000), Wasserstein Distance (Earth Mover's Distance) Rubner et al. (2000), and Hellinger Distance Thomas and Joy (2006).

Random Sampling l options

We then sample l at-the-money (ATM) or out-of-the-money (OTM) options from the N simulated options with replacement, ensuring that $l \ll N$. Subsequently, we employ CGP and other methods to estimate the risk-neutral densities p_{CGP} , p_{GP} , and others, and proceed to compare them.

We can assess how the differences in the risk-neutral densities vary with changes in the values of k and l . This enables us to address the following questions:

1. Whether p_{CGP} can be consistently and accurately extracted under varying volatility and other simulation parameters.
2. Whether p_{CGP} demonstrates greater accuracy under specific conditions, such as when only ATM options are used.
3. Whether p_{CGP} can still be accurately extracted with small values of k and l .

Comparing RND

The L^2 distance (or Euclidean distance) between the risk-neutral densities (RNDs) provides a quantitative measure of the differences between them. The L^2 distance is defined as:

$$L^2(p, q) = \sqrt{\int (p(x) - q(x))^2 dx}$$

where $p(x)$ and $q(x)$ are the two densities being compared. This method is straightforward and provides an intuitive measure of the overall difference between the two densities.

Kullback-Leibler (KL) Divergence Kullback and Leibler (1951):

- The KL divergence measures the difference between two probability distributions. It is defined

as:

$$D_{\text{KL}}(p \parallel q) = \int p(x) \log \left(\frac{p(x)}{q(x)} \right) dx$$

- The KL divergence is not symmetric, meaning $D_{\text{KL}}(p \parallel q) \neq D_{\text{KL}}(q \parallel p)$. It provides a measure of how much information is lost when q is used to approximate p .

Jensen-Shannon (JS) Divergence Tishby et al. (2000):

- The JS divergence is a symmetric measure based on the KL divergence. It is defined as:

$$D_{\text{JS}}(p \parallel q) = \frac{1}{2} D_{\text{KL}}(p \parallel m) + \frac{1}{2} D_{\text{KL}}(q \parallel m)$$

where $m = \frac{1}{2}(p + q)$.

- The JS divergence is bounded and provides a more interpretable measure compared to the KL divergence.

Wasserstein Distance (Earth Mover's Distance) Rubner et al. (2000): - The Wasserstein distance measures the cost of transforming one distribution into another. For 1D distributions, it is defined as:

$$W(p, q) = \int |P(x) - Q(x)| dx$$

where P and Q are the cumulative distribution functions (CDFs) of p and q .

Hellinger Distance Thomas and Joy (2006): - The Hellinger distance is a measure of the similarity between two probability distributions. It is defined as:

$$H(p, q) = \frac{1}{\sqrt{2}} \sqrt{\int (\sqrt{p(x)} - \sqrt{q(x)})^2 dx}$$

- The Hellinger distance ranges from 0 to 1, where 0 indicates identical distributions and 1 indicates maximal difference.

Calculating the L^2 distance between the theoretical RND $p(H)$ and the estimated RNDs is a sensible approach to measure differences. Additionally, using other metrics like the KL divergence, JS divergence, Wasserstein distance, and Hellinger distance can provide further insights into the nature of the differences between the distributions. Each of these methods has its own advantages and can be used to complement the L^2 distance in a comprehensive analysis.

References

- Heston, S.L., 1993. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The review of financial studies* 6, 327–343.
- Kullback, S., Leibler, R.A., 1951. On information and sufficiency. *The annals of mathematical statistics* 22, 79–86.
- Rubner, Y., Tomasi, C., Guibas, L.J., 2000. The earth mover’s distance as a metric for image retrieval. *International journal of computer vision* 40, 99–121.
- Thomas, M., Joy, A.T., 2006. *Elements of information theory*. Wiley-Interscience.
- Tishby, N., Pereira, F.C., Bialek, W., 2000. The information bottleneck method. *arXiv preprint physics/0004057* .